

current increases. Hence, the separation of the current scale will decrease as the current increases.

3 C Constant acceleration a

$$\text{So } v^2 = u^2 + 2as \Rightarrow as = \frac{1}{2} (v^2 - u^2)$$

$$\text{At Y, } v_Y^2 = u^2 + 2a(s/2) = u^2 + as = u^2 + \frac{1}{2} (v^2 - u^2) = \frac{1}{2} (u^2 + v^2)$$

4 C During fall in air, weight of ball - viscous force = ma